Zewail City of Science, Technology and Innovation
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School of Computational Sciences and Artificial Intelligence

CSAI 203 – Fall 2026

Introduction to Software Engineering
Academic Course & Schedule Planner

Software Requirements Specification

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Outline

1. Introduction....3
1.1 Purpose....3
1.2 Scope....3
1.3 Definitions, Acronyms, and Abbreviations....3
1.4 References....3
1.5 Overview....4
2. Overall Description....4
2.1 Product Perspective....4
2.2 Product Functions....4
2.3 User Classes and Characteristics....4
2.4 Software Stakeholders....5
2.5 Operating Environment....5
2.6 Design and Implementation Constraints....5
2.7 User Documentation....5
2.8 Assumptions and Dependencies....5
3.0 Specific Requirements....6
3.1 Functional Requirements....6
3.2 Use Case Model....7
3.2.1 Use Case Diagrams....7
3.2.2 Use Case Descriptions....9
3.3 Domain Model....14
3.3.1 Conceptual Class Diagram....14
3.3.2 Class Descriptions....14
3.4 Non-Functional Requirements....15
3.5 External Interface Requirements....16
3.5.1 User Interface....16
3.5.2 Hardware Interface....16
3.5.3 Software Interface....16
3.5.4 Communication Interface....16
4. Appendices....17
4.1 Appendix A: Data Dictionary....17
4.2 Appendix B: Glossary....17

1. Introduction

1.1 Purpose

The purpose of this document is to specify the functional and non-functional requirements of the Academic Course & Schedule Planner. It defines the system's scope, intended users, features, and constraints, and serves as a reference for the development team, the course instructors/TAs, and the project stakeholders. It is the baseline against which later design, implementation, and testing are evaluated.

1.2 Scope

The Academic Course & Schedule Planner is a web app to help students build timetables per semester with features including conflict checking, tracking prerequisite requirements, GPA calculation, visual timetable, and automatic best-case schedule generation for the students. The app also allows admins to manage course offerings, capacities, and prerequisites.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
SRS	Software Requirements Specification - this document.
FR	Functional Requirement -- a feature or behavior the system must perform.
NFR	Non-Functional Requirement -- a quality attribute or constraint on the system.
UC	Use Case - a sequence of interactions between an actor and the system that yields a result of value.
UML	Unified Modeling Language -- the standard visual modeling notation used for the diagrams in this document.
MVC	Model-View-Controller -- the mandatory architectural pattern for this project, separating data (Model), interface (View), and input/coordination (Controller).
GPA	Grade Point Average -- the numeric average of a student's course grades.
Credit Hour	The unit measuring a course's academic weight.
Prerequisite	A course that must be completed before another course can be taken.
Section	A specific offering of a course in a semester, with its own time slot and capacity.
Timetable	A weekly view showing when each planned course section meets.
Actor	A person or external system that interacts with the system (e.g., Student, Administrator).
Admin	Administrator -- the user role that manages courses, sections, and prerequisites.

1.4 References

1] Object Management Group. (2017). OMG Unified Modeling Language (UML) Specification, Version 2.5.1. https://www.omg.org/spec/UML/2.5.1/

[2] OWASP Foundation. (2021). OWASP Top 10:2021 - The Ten Most Critical Web Application Security Risks.https://owasp.org/Top10/

[3] Pallets Projects. (2024). Flask Documentation. https://flask.palletsprojects.com/

1.5 Overview

The document is organised into three sections. Section 2 contains an overall description and the characteristics of the project including the users, functions, operating environment, design and user documentation. Section 3 specifies the functional and non-functional requirements, use cases, domain model, and external interface. Section 4 contains the appendices inclduing the data dictionary and glossary.

2. Overall Description

2.1 Product Perspective

The Academic Course & Schedule Planner is a standalone web application to assist with planning student's course schedules according to the available sections and courses managed by the admins. It is intended to be used alongisde the university's existing systems rather than replace them.

2.2 Product Functions

The Academic Course & Schedule provides the following main functionality:

- Course and section management: Allows adminstrators to manage course offered each semester including the capacities, sections, and prerequisites.
- Schedule planning: Allows students to plan their schedules based on the courses available managed by the adminstrators. Students can add, remove, and search courses and check their prerequisites.
- Schedule conflict checking: Schedules have checking which lets the student know if courses cause a time conflict between each other.
- Schedule prerequisite checking: The system will alert the student if the student doesn't meet a prerequisite requirement.
- GPA calculation: Easy to use tool within the app to calculate GPA for each semester.
- Course searching and filtering: The student can use the search tool to search for courses and filter based on major, prerequisites, credit hours, etc.

2.3 User Classes and Characteristics

- **Students**

Students are the primary end users of the application. They use it to manage and plan their schedules to help them plan ahead and identify scheduling conflicts, check/plan for course prerequisites, and track their academic progress. They are expected to have basic computer knowledge and have a modern browser.

- **Adminstrators**

Adminstrators mainly manage the academic information and data in the system and updating it whenever schedules and sections change, creating and updating course sections, defining course prerequisites, credit hours, and course offering each semester. They are expected to have knowledge of the university academic structure and course system.

2.4 Software Stakeholders

- **Students**

The primary stakeholders who benefit from simplified course scheduling, prerequisite checking, and academic progress tracking.

- **University Adminstration**

The system can ease registeration periods. When students have a plan they can efficeintly choose courses to register quickly and reduce load on the servers and registeration confusion.

2.5 Operating Environment

The Academic Course & Schedule Planner is a web app accessible through modern browsers such as firefox (gecko based) and chrome (chromium based) browsers. Users must have an internet connection to access the application.

The system will run on a web server to store and handle data. The user client will interact with the application through the web application.

The application should work on any operating system including Windows, Linux, MacOS, Android, IOS provided the operating system can run a modern browser.

2.6 Design and Implementation Constraints

The project must be implemented in python as required by the course.

2.7 User Documentation

The system will provide a Frequently Asked Questions (FAQ) section to help users understand its features and resolve common issues.

2.8 Assumptions and Dependencies

The web application assumes that users have a stable internet connection and an up-to-date web browser. It also assumes that administrators maintain accurate and reliable information about courses, prerequisites, credit hours, and course sections.

3.0 Specific Requirements

3.1 Functional Requirements

FR01 - Student Registration: The system shall allow students to create an account with a unique username and a password, rejecting any username that already exists.

FR02 - User Login: The system shall allow students and admins to log in using their registered username and password and verify the credentials against the stored data.

FR03 - Create Courses: The system shall allow admins to create courses by specifying the course ID, name, description, major, and credit hours.

FR04 - Edit Course Information: The system shall allow admins to edit existing course information to keep it accurate and up to date.

FR05 - Manage Course Prerequisites: The system shall allow admins to add, modify, and remove prerequisites for a course and shall validate that each referenced prerequisite course exists.

FR06 - Manage Semester Offerings, Sections, and Capacities: The system shall allow admins to define course offerings and sections for a semester (including multiple sections and time slots per course) and to set and update the capacity of each section.

FR07 - Browse Course Catalog: The system shall allow students to browse the full catalog of courses offered by the university.

FR08 - View Course Details: The system shall allow students to view a course's details, including prerequisites, credit hours, and available sections for the current semester.

FR09 - Search and Filter Courses: The system shall allow students to search courses by ID or name and filter them by department and by prerequisite eligibility.

FR10 - Check Prerequisite Eligibility: The system shall check whether a student meets a course's prerequisites and warn the student when attempting to add a course they are not eligible for.

FR11 - Record Completed Courses and Grades: The system shall record students' completed courses and grades and allow students to manually add completed courses and their grades.

FR12 - Calculate Semester GPA: The system shall calculate the student's current GPA from the recorded completed courses and grades.

FR13 - Create Schedule Plan Manually: The system shall allow students to manually build a semester schedule from the available course offerings and sections.

FR14 - Generate Automatic Best-Case Schedule: The system shall generate an automatic schedule from the student's planned courses, selecting section times that minimize gaps between lectures.

FR15 - Remove Courses or Sections from a Plan: The system shall allow students to remove courses or sections from their plan and modify the plan as needed.

FR16 - Detect Schedule Conflicts: The system shall detect time conflicts between selected course sections and warn the student.

FR17 - View Planned Schedule Timetable: The system shall display an interactive weekly timetable showing each planned section's start and end times.

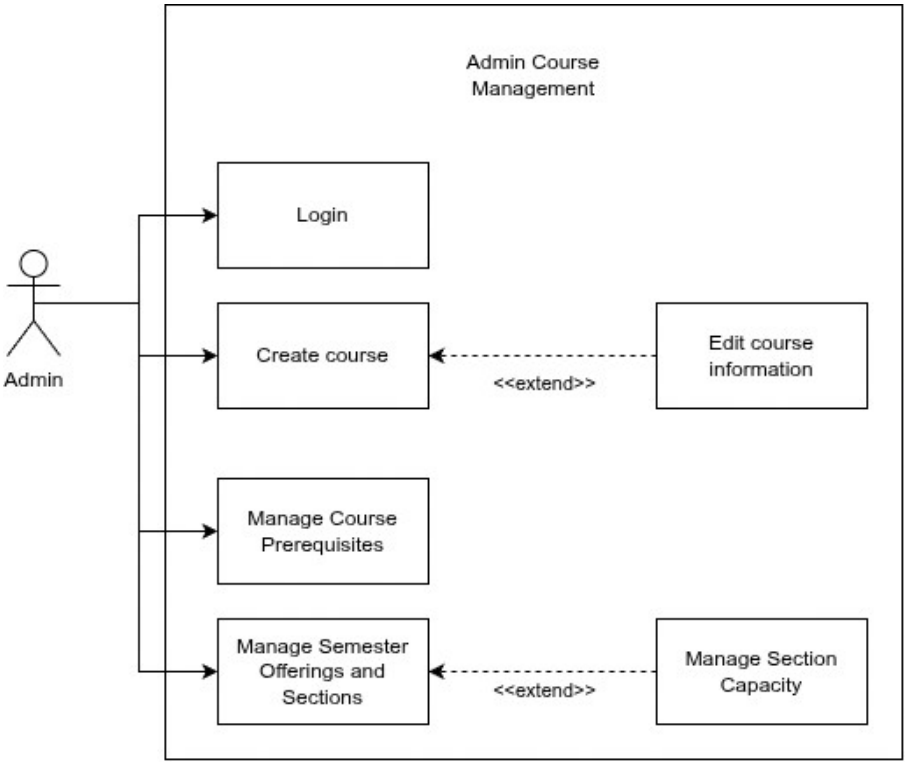
FR18 - Track Completed and Planned Credit Hours: The system shall track and display the student's completed and planned credit hours.

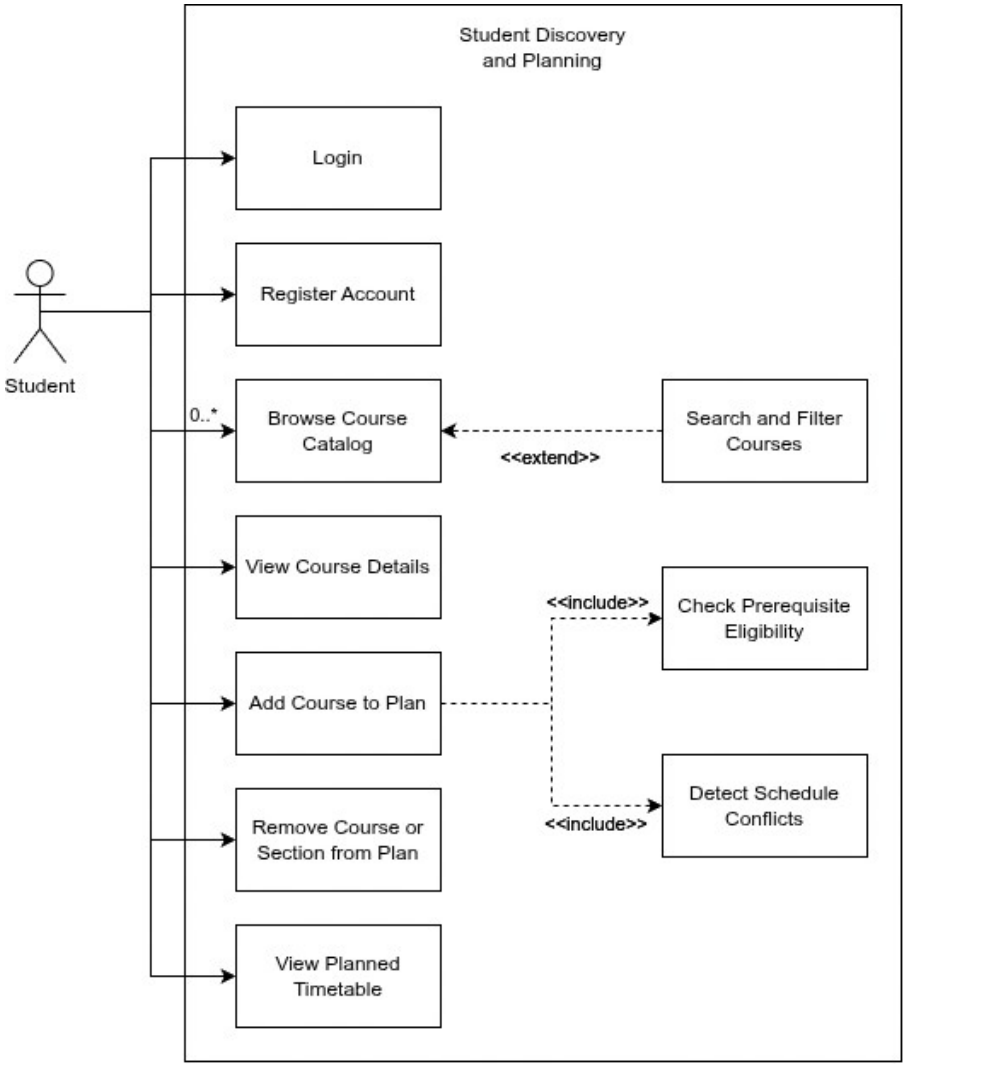
FR19 - Warn About Credit-Hour Limits: The system shall warn the student when planning a schedule that exceeds the allowed credit-hour limit.

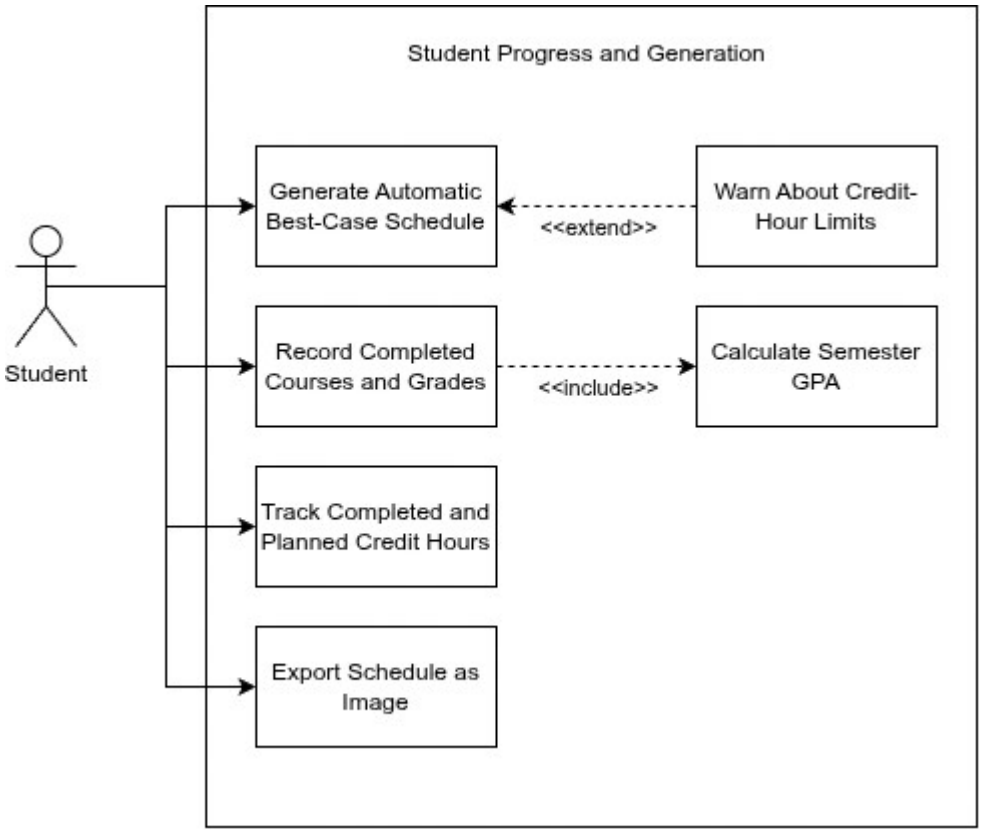
FR20 - Export Schedule as Image: The system shall allow students to export their schedule as an image for easy sharing.

3.2 Use Case Model

3.2.1 Use Case Diagrams




Student Discovery and Planning
Login
Register Account
Student
0..*
Browse Course Catalog
Search and Filter Courses
<<extend>>
View Course Details
<<include>>
Check Prerequisite Eligibility
Add Course to Plan
Detect Schedule Conflicts
<<include>>
Remove Course or Section from Plan
View Planned Timetable


Student Progress and Generation
Generate Automatic Best-Case Schedule
Warn About Credit-Hour Limits
<<extend>>
Student
Record Completed Courses and Grades
Calculate Semester GPA
<<include>>
Track Completed and Planned Credit Hours
Export Schedule as Image

3.2.2 Use Case Descriptions

ID	Name	Description	Actors	Precondition	Postcondition	Break Condition	Main flow
UC-01	Login	An existing student or administrator logs into the system using their registered username and password.	Student, Administrator	The user has a registered account.	The user is authenticated and redirected to their respective dashboard.	The username does not exist, or the password is incorrect.	1. User opens the login page. 2. User enters username and password. 3. System validates the credentials against stored data. 4. System grants access and redirects to the appropriate dashboard.
UC-02	Create Course	An administrator adds a new course to the system by specifying its core academic details.	Administrator	Administrator is logged in.	The new course is saved in the catalog and visible to students.	The course ID already exists, or required fields are missing/invalid.	1. Administrator opens the course management page. 2. Administrator enters course ID, name, description, major, and credit hours. 3. System validates the input and checks that the course ID is unique. 4. System saves the new course. 5. System displays a success message.
UC-03	Edit Course Information	An administrator modifies the details of an existing course to correct mistakes or keep information current.	Administrator	Administrator is logged in and the course exists.	The updated course information is saved and reflects in the catalog.	The course is not found, or the new data is invalid.	1. Administrator searches for and selects an existing course. 2. Administrator edits the desired fields. 3. System validates the new input. 4. System saves the updated course record. 5. System displays a success message.
UC-04	Manage Course Prerequisites	An administrator defines, modifies, or removes the prerequisite courses required for a	Administrator	Administrator is logged in and the target course exists.	The course's prerequisite list is updated and saved.	A referenced prerequisite course does not exist in the system.	1. Administrator selects a course. 2. Administrator adds, modifies, or removes prerequisite entries. 3. System

		specific course.					validates that each referenced prerequisite course exists. 4. System saves the updated prerequisite list.
UC-05	Manage Semester Offerings and Sections	An administrator defines which courses are offered in a semester and creates their specific sections and time slots.	Administrator	Administrator is logged in and the relevant courses exist.	The semester offering and its sections are saved and available for student planning.	The selected course does not exist, or the section time slot is invalid.	1. Administrator selects a semester and a course. 2. Administrator defines one or more sections with their time slots. 3. System validates the course exists and the times are well-formed. 4. System saves the offerings and sections.
UC-06	Manage Section Capacity	An administrator sets or updates the maximum number of students allowed in a course section.	Administrator	Administrator is logged in and the section exists.	The section's capacity is updated and visible to students.	The entered capacity is negative or below the number of already-enrolled students.	1. Administrator selects a section. 2. Administrator enters a new capacity value. 3. System validates the value. 4. System saves the updated capacity.
UC-07	Register Account	A new student creates an account with a unique username and password to access the planner.	Student	The user is not already logged in.	A new student account is created and the user is redirected to login.	The chosen username already exists, or the password does not meet the minimum rules.	1. Student opens the registration page. 2. Student enters username, password, and confirms the password. 3. System validates the input (uniqueness, password rules). 4. System stores the new account. 5. System displays a success message and redirects to login.
UC-08	Browse Course Catalog	A student views the full list of courses offered by the university.	Student	Student is logged in.	The course catalog is displayed to the student.	The catalog is empty or the server fails to retrieve it.	1. Student navigates to the course catalog page. 2. System retrieves the list of

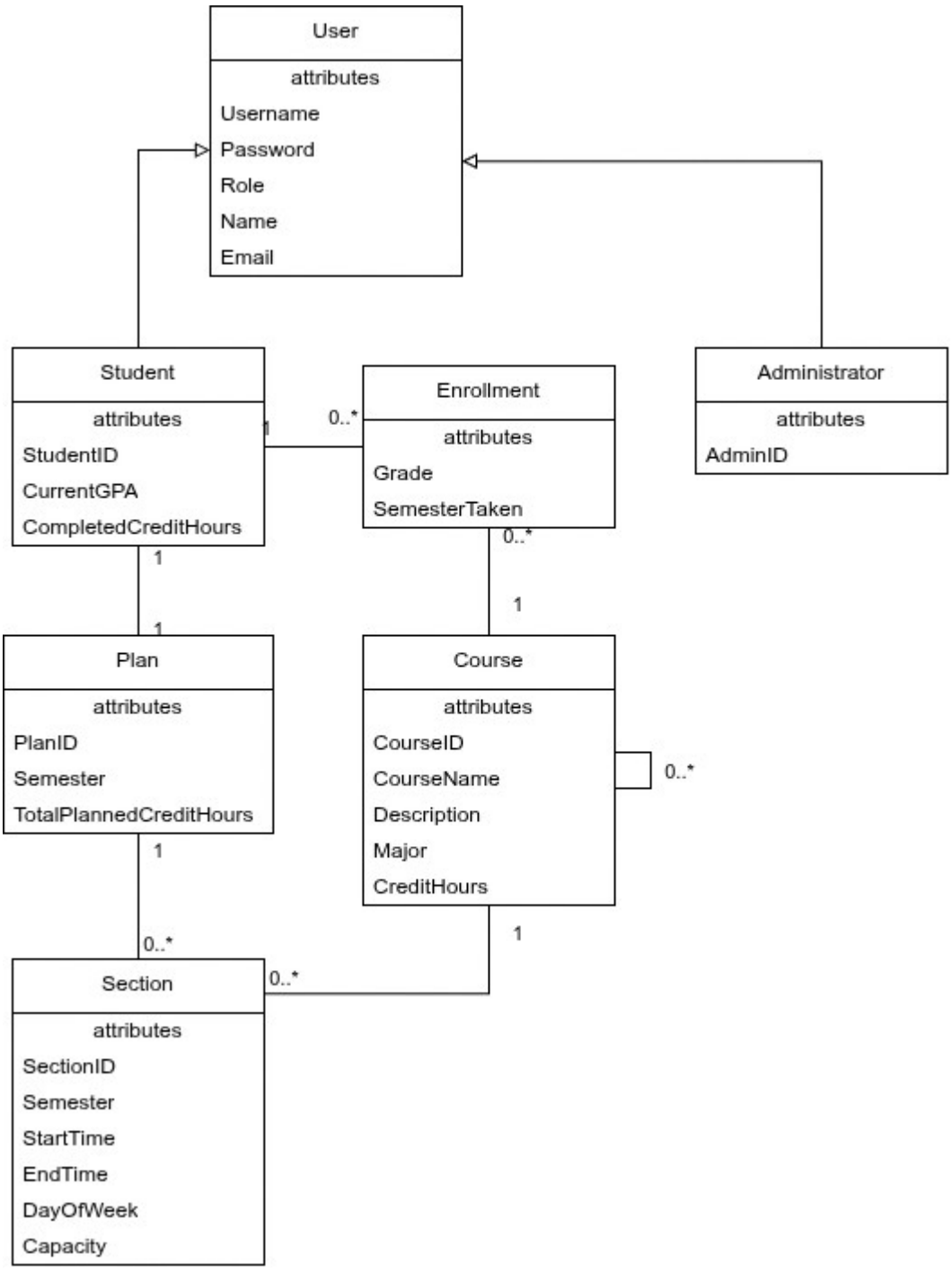
							courses. 3. System displays the courses to the student.
UC-09	View Course Details	A student views detailed information about a selected course, including prerequisites, credit hours, and available sections.	Student	Student is logged in and has selected a course from the catalog.	The detailed course information is displayed.	The selected course is no longer available.	1. Student selects a course from the catalog. 2. System retrieves the course details and available sections. 3. System displays the details to the student.
UC-10	Search and Filter Courses	A student searches for courses by ID or name and filters the results by department and prerequisite eligibility.	Student	Student is logged in and viewing the catalog.	A filtered list of matching courses is displayed.	No courses match the search/filter criteria (system displays an empty-state message).	Main flow 1. Student enters a search term and/or selects filter criteria. 2. System queries the catalog based on the criteria. 3. System displays the matching results.
UC-11	Add Course to Plan	A student adds a specific course section to their semester plan after the system verifies eligibility and checks for conflicts.	Student	Student is logged in and at least one section exists for the selected course.	The section is added to the student's plan, or the addition is rejected with a clear error message.	The student does not meet the prerequisites, or the section conflicts with an already-planned section.	1. Student selects a section to add. 2. System checks prerequisites (includes UC-12). 3. System checks for time conflicts (includes UC-13). 4. System adds the section to the plan. 5. System displays a confirmation message.
UC-12	Check Prerequisite Eligibility	The system verifies whether a student has completed all prerequisites required for a selected course.	Student	Precondition The student's completed courses are recorded and the target course has a defined prerequisite list.	The system determines eligibility and informs the student if they are ineligible.	The student's academic record is incomplete or missing.	1. System retrieves the course's prerequisite list. 2. System retrieves the student's completed courses. 3. System compares the two sets. 4. System returns an eligibility result (met/not met).
UC-13	Detect Schedule Conflicts	The system checks whether a newly	Student	The student's plan contains at least one	The system reports whether a conflict	The section times are malformed or	1. System retrieves the times of the selected

		selected section overlaps in time with any section already in the student's plan.		section (or the check confirms no overlap).	exists.	missing.	section. 2. System retrieves the times of all planned sections. 3. System compares the time intervals. 4. System returns a conflict result (conflict/no conflict).
UC-14	Remove Course or Section from Plan	A student removes a course or section from their current semester plan.	Student	Student is logged in and the plan contains at least one section.	The selected course/section is removed from the plan.	The selected course/section is not in the plan.	1. Student selects a course/section in the plan. 2. Student confirms removal. 3. System removes the item from the plan. 4. System updates the timetable and credit-hour count.
UC-15	View Planned Timetable	A student views an interactive weekly timetable showing when each planned section starts and ends.	Student	Student is logged in and has at least one section in the plan.	The timetable is displayed.	The plan is empty (system displays an empty-state message).	1. Student opens the timetable view. 2. System retrieves the planned sections and their times. 3. System renders the weekly timetable.
UC-16	Generate Automatic Best-Case Schedule	The system automatically generates a schedule from the student's selected courses, choosing section times that minimize gaps between lectures.	Student	Student is logged in and has selected one or more courses to schedule.	A generated schedule is presented to the student for acceptance.	No valid combination of sections exists (e.g., unavoidable conflicts).	1. Student selects the courses to schedule. 2. System retrieves all available sections for those courses. 3. System evaluates combinations and selects the one with the smallest gaps. 4. System presents the generated schedule.
UC-17	Record Completed Courses and Grades	A student records completed courses and their grades manually so the system can compute GPA and check	Student	Student is logged in.	The completed course and grade are stored; GPA is recalculated (includes UC-18).	The course does not exist or the grade is invalid.	1. Student selects a completed course. 2. Student enters the grade. 3. System validates the course and grade. 4. System stores

		prerequisites.					the record. 5. System recalculates GPA (includes UC-18).
UC-18	Calculate Semester GPA	The system calculates the student's GPA from recorded completed courses and grades using a 4.0 scale.	Student	At least one completed course with a grade is recorded.	The updated GPA is displayed and stored.	No graded courses exist (GPA is not computed).	1. System retrieves all recorded graded courses. 2. System maps each grade to its grade points (4.0 scale). 3. System computes the weighted average. 4. System stores and displays the GPA.
UC-19	Track Completed and Planned Credit Hours	The system tracks and displays the student's completed credit hours and the credit hours planned for the current semester.	Student	Student is logged in and has completed courses and/or a current plan.	The completed and planned credit-hour totals are displayed.	No data is available (both totals show as zero).	1. System retrieves completed courses and their credit hours. 2. System retrieves planned sections and their credit hours. 3. System sums each total. 4. System displays both totals.
UC-20	Warn About Credit-Hour Limits	The system warns the student when the planned credit hours exceed the allowed limit for the semester.	Student	The student has at least one section in the plan.	The student is warned and may adjust the plan or confirm continuing.	The planned credit hours are within the allowed limit (no warning is displayed).	1. System calculates total planned credit hours. 2. System compares the total to the allowed limit. 3. System displays a warning if the total exceeds the limit.
UC-21	Export Schedule as Image	A student exports their planned schedule as an image file for easy sharing.	Student	Student is logged in and the plan is not empty.	An image file of the schedule is generated and downloaded.	The plan is empty or the image generation fails.	1. Student clicks the export button. 2. System renders the timetable as an image. 3. System provides the image file for download.

3.3 Domain Model

3.3.1 Conceptual Class Diagram



3.3.2 Class Descriptions

1. User

Purpose: Represents a generic account in the system, handling authentication and role assignment.

Attributes: Username, Password, Role, Name, Email

2. Student

Purpose: Represents a student who plans schedules, checks prerequisites, and tracks academic progress.

Attributes: StudentID, CurrentGPA, CompletedCreditHours

3. Administrator

Purpose: Represents a university staff member who manages courses, sections, and prerequisites.

Attributes: AdminID

4. Course

Purpose: Represents a real-world university course offered in the catalog.

Attributes: CourseID, CourseName, Description, Major, CreditHours

5. Section

Purpose: Represents a specific offering of a course in a semester, including its schedule and capacity.

Attributes: SectionID, Semester, StartTime, EndTime, DayOfWeek, Capacity

6. Plan

Purpose: Represents a student's draft schedule for the current semester.

Attributes: PlanID, Semester, TotalPlannedCreditHours

7. Enrollment

Purpose: Represents a completed course and the grade the student received.

Attributes: Grade, SemesterTaken

3.4 Non-Functional Requirements

ID	Requirement	Test Plan	Success Criteria
NFR-01	Performance: The system shall load the course catalog and generate an automatic schedule within 2 seconds under normal network conditions.	Measure the time from clicking "Generate" to full render using browser DevTools.	≤ 2000ms for 95% of requests
NFR-02	Security: The system shall enforce role-based access (Students cannot access Admin pages) and store passwords using secure hashing (e.g., bcrypt).	1. Log in as Student, try to access Admin URL. 2. Inspect DB password field.	Student is blocked (403 error); no plain-text passwords in DB.
NFR-03	Usability: The UI shall be responsive and usable on desktop and mobile screens (min. width 320px) with clear navigation.	Test on mobile, attempt to add a course.	No horizontal scrolling, all buttons clickable, text readable without zoom.
NFR-04	Reliability: The system shall handle DB/network errors gracefully without crashing and not save partial plans if a conflict is detected.	Disconnect DB while clicking "Add Course."	Friendly error shown, system doesn't crash, plan remains unchanged.
NFR-05	Portability: The web app shall function consistently on Chromium (Chrome/Edge) and Gecko (Firefox) browsers.	Run primary flows (Login, Browse, Plan, Generate) on Chrome, Edge, Firefox.	All features work, UI layout is consistent, no broken elements.

3.5 External Interface Requirements

3.5.1 User Interface

The system provides a web-based graphical user interface (GUI) accessible through modern web browsers. The UI is divided into distinct views based on the user's role:

- **Student Portal:** Includes the course catalog, search/filter tools, the interactive weekly timetable, the schedule generation tool, and the GPA/credit hour tracker.
- **Admin Dashboard:** Includes forms for creating and editing courses, managing prerequisites, defining semester sections, and setting section capacities.
- **General:** The interface includes a login/registration page and a navigation bar.

3.5.2 Hardware Interface

There are no direct hardware interfaces required.

3.5.3 Software Interface

- Web Server: The application is hosted on a web server that handles HTTP/HTTPS requests and serves the application to the client.
- Backend Framework: The backend logic is implemented in Python (e.g., using Flask), following the mandatory MVC architectural pattern.

3.5.4 Communication Interface

All communication between the client browser and the server is conducted over HTTPS to ensure secure and encrypted data transfer.

4. Appendices

4.1 Appendix A: Data Dictionary

Data Entity	Description	Key Attributes
User	Stores all accounts (Students and Admins).	UserID, Username, PasswordHash, Role, GPA
Course	The master list of all university courses.	CourseID, Name, Description, Major, CreditHours
Prerequisite	Links courses that require prior completion.	CourseID, PrerequisiteCourseID
Section	Specific semester offerings of a course.	SectionID, CourseID, Semester, TimeSlot, Capacity
Enrollment	Tracks completed courses and grades.	StudentID, CourseID, Grade, SemesterTaken
Plan	The student's active semester schedule, containing the selected sections.	PlanID, StudentID, Semester, SelectedSections

4.2 Appendix B: Glossary

Term	Definition
SRS	Software Requirements Specification -- this document.
FR / NFR	Functional Requirement / Non-Functional Requirement.
UML	Unified Modeling Language -- the standard notation used for the diagrams in this document.
MVC	Model-View-Controller -- the mandatory architectural pattern separating data, interface, and logic.
GPA	Grade Point Average -- calculated on a 4.0 scale.
Credit Hour	The unit of academic weight assigned to a course.
Prerequisite	A required prior course that must be passed before enrolling in a specific course.
Section	A scheduled instance of a course (e.g., a specific day/time slot).
Timetable	A weekly visual representation of a student's planned sections.
Actor	A person (Student, Admin) who interacts with the system.
Break Condition	An event or state that causes a use case's main flow to fail or take an alternative path.
Traceability	The ability to link requirements to design, code, and tests.